

Computer Networks: Assignment-1

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Q 1) Study network configuration commands and execute them on your system. For Example:

1. IPCONFIG

Command:

ipconfig

Output:

```
Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::b412:719e:b04a:19e6%17
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : mshome.net
    IPv4 Address. . . . . : 192.168.55.42
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.55.254

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

2. NSLOOKUP

Command:

nslookup -type = google.com

Output:

```
DNS request timed out.  
    timeout was 2 seconds.  
Server:  UnKnown  
Address: 172.217.174.78  
  
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
*** Request to UnKnown timed-out
```

3. HOSTNAME

Command:

hostname

Output:

```
LAPTOP-484SUN9N
```

4. PING

Command:

ping google.com

Output:

```
Reply from 142.250.67.238: bytes=32 time=54ms TTL=112  
Reply from 142.250.67.238: bytes=32 time=39ms TTL=112  
Reply from 142.250.67.238: bytes=32 time=41ms TTL=112  
Reply from 142.250.67.238: bytes=32 time=46ms TTL=112  
  
Ping statistics for 142.250.67.238:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 39ms, Maximum = 54ms, Average = 45ms
```

5. TRACERT

Command:

```
tracert google.com
```

Output:

```
Tracing route to google.com [142.250.67.238]
over a maximum of 30 hops:

  1      8 ms      13 ms      2 ms  192.168.55.254
  2     25 ms      6 ms      9 ms  192.0.0.1
  3      *         *         *    Request timed out.
  4      *         *         *    Request timed out.
  5      *         *         *    Request timed out.
  6      *         *         *    Request timed out.
  7      *         *         *    Request timed out.
  8      *         *         *    Request timed out.
  9      *         *         *    Request timed out.
 10     *         *         *    Request timed out.
 11     *         *         *    Request timed out.
 12     *         *         *    Request timed out.
 13     *         *         *    Request timed out.
 14    40 ms     34 ms     32 ms  bom07s24-in-f14.1e100.net [142.250.67.238]

Trace complete.
```

6. NETSTAT

Command:

```
netstat
```

Output:

```
Active Connections

 Proto Local Address           Foreign Address         State
TCP    192.168.55.42:49410      4.213.25.240:https      ESTABLISHED
TCP    192.168.55.42:51094      whatsapp-cdn-shv-04-bom2:https ESTABLISHED
TCP    192.168.55.42:53337      bom12s17-in-f10:https   TIME_WAIT
TCP    192.168.55.42:55140      ec2-52-7-5-165:https    CLOSE_WAIT
TCP    192.168.55.42:56436      whatsapp-cdn-shv-04-bom2:5222 ESTABLISHED
TCP    192.168.55.42:57744      104.18.39.21:https      ESTABLISHED
TCP    192.168.55.42:57745      52.123.129.14:https     ESTABLISHED
TCP    192.168.55.42:57747      20.106.86.13:https      TIME_WAIT
TCP    192.168.55.42:57748      150.171.27.12:https     ESTABLISHED
TCP    192.168.55.42:57749      49.44.138.145:https     ESTABLISHED
TCP    192.168.55.42:58368      104.18.32.47:https      ESTABLISHED
TCP    192.168.55.42:58714      172.64.155.209:https    ESTABLISHED
TCP    192.168.55.42:59350      se-in-f188:https        ESTABLISHED
TCP    192.168.55.42:60139      4.213.25.240:https      ESTABLISHED
TCP    192.168.55.42:60411      4.186.14.122:https      ESTABLISHED
TCP    192.168.55.42:65231      151.101.209.55:https    ESTABLISHED
TCP    192.168.55.42:65234      ec2-34-216-166-254:https CLOSE_WAIT
TCP    192.168.55.42:65236      ec2-52-1-166-83:https   TIME_WAIT
```

7. ARP(Address Resolution Protocol)

Command:

arp -a

Output:

Active Connections			
Proto	Local Address	Foreign Address	State
TCP	192.168.55.42:49410	4.213.25.240:https	ESTABLISHED
TCP	192.168.55.42:51094	whatsapp-cdn-shv-04-bom2:https	ESTABLISHED
TCP	192.168.55.42:53337	bom12s17-in-f10:https	TIME_WAIT
TCP	192.168.55.42:55140	ec2-52-7-5-165:https	CLOSE_WAIT
TCP	192.168.55.42:56436	whatsapp-cdn-shv-04-bom2:5222	ESTABLISHED
TCP	192.168.55.42:57744	104.18.39.21:https	ESTABLISHED
TCP	192.168.55.42:57745	52.123.129.14:https	ESTABLISHED
TCP	192.168.55.42:57747	20.106.86.13:https	TIME_WAIT
TCP	192.168.55.42:57748	150.171.27.12:https	ESTABLISHED
TCP	192.168.55.42:57749	49.44.138.145:https	ESTABLISHED
TCP	192.168.55.42:58368	104.18.32.47:https	ESTABLISHED
TCP	192.168.55.42:58714	172.64.155.209:https	ESTABLISHED
TCP	192.168.55.42:59350	se-in-f188:https	ESTABLISHED
TCP	192.168.55.42:60139	4.213.25.240:https	ESTABLISHED
TCP	192.168.55.42:60411	4.186.14.122:https	ESTABLISHED
TCP	192.168.55.42:65231	151.101.209.55:https	ESTABLISHED
TCP	192.168.55.42:65234	ec2-34-216-166-254:https	CLOSE_WAIT
TCP	192.168.55.42:65236	ec2-52-1-166-83:https	TIME_WAIT

8. SYSTEMINFO

Command:

systeminfo

Output:

```
Host Name: LAPTOP-484SUN9N
OS Name: Microsoft Windows 11 Home Single Language
OS Version: 10.0.26100 N/A Build 26100
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Workstation
OS Build Type: Multiprocessor Free
Registered Owner: Suraj Shah
Registered Organization: HP
Product ID: 00342-42714-19069-AAOEM
Original Install Date: 25-03-2025, 08:35:51
System Boot Time: 09-01-2026, 12:21:32
System Manufacturer: HP
System Model: HP Laptop 15-fd1xxx
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: Intel64 Family 6 Model 170 Stepping 4 GenuineIntel ~1200 Mhz
BIOS Version: AMI F.07, 20-08-2024
Windows Directory: C:\WINDOWS
System Directory: C:\WINDOWS\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: 00004009
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 15,801 MB
Available Physical Memory: 5,272 MB
Virtual Memory: Max Size: 24,505 MB
Virtual Memory: Available: 13,173 MB
Virtual Memory: In Use: 11,332 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \\LAPTOP-484SUN9N
Hotfix(s): 3 Hotfix(s) Installed.
[01]: KB5066131
[02]: KB5072033
[03]: KB5071142
Network Card(s): 3 NIC(s) Installed.
[01]: Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Connection Name: Wi-Fi
DHCP Enabled: Yes
DHCP Server: 192.168.55.254
IP address(es)
```

```
[01]: 192.168.56.1
[02]: fe80::b412:719e:b04a:19e6
Virtualization-based security: Status: Running
Required Security Properties:
Available Security Properties:
Base Virtualization Support
Secure Boot
DMA Protection
UEFI Code Readonly
SMM Security Mitigations 1.0
Mode Based Execution Control
APIC Virtualization
Services Configured:
Hypervisor enforced Code Integrity
Services Running:
Hypervisor enforced Code Integrity
App Control for Business policy: Enforced
App Control for Business user mode policy: Off
Security Features Enabled:
Hyper-V Requirements: A hypervisor has been detected. Features required for Hyper-V will not be displayed.
```